

**MATHEMATICS**  
**Modern Algebra I**  
Suggested exercises 1

- (1) Let  $G$  be a group such that  $x^2 = 1$  for any  $x \in G$ . Prove that  $G$  is abelian.
- (2) Show that any cyclic group is isomorphic either to  $\mathbb{Z}$  or to  $\mathbb{Z}/m\mathbb{Z}$ .
- (3) Describe all subgroups of a cyclic group.
- (4) Let  $G$  be a finite group, show that  $x^{|G|} = 1$  for all  $x \in G$ .
- (5) Show that if  $G$  is a finite group of prime order, then  $G$  is cyclic.
- (6) Show that  $\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/nm\mathbb{Z}$  if and only if  $n, m$  are relatively prime.
- (7) Let  $\phi : G \rightarrow G'$  be an homomorphism of groups,  $N := \phi^{-1}(N')$ . Show that if  $N'$  is a normal subgroup of  $G'$ , then  $N$  is a normal subgroup of  $G$ .
- (8) If  $H \subset G$  is a subgroup of  $G$  of index 2, show that  $H$  is normal in  $G$ .
- (9) Find all possible orders of the elements of  $S_7$ .
- (10) Let  $H \subset G$  be a subset of a group. Show that  $x \sim y$  iff  $xy^{-1} \in H$  is an equivalence relation iff  $H$  is a subgroup.
- (11) Let  $H \subset G$  be a subgroup, show that the multiplication on  $G$  induces a group structure on  $G/H$  iff  $H$  is normal in  $G$ .

**The midterms will be on Wednesday October 13 and Friday November 19.**